Population

The big idea

People are the most important element of a country: they produce and consume resources, and they are themselves a resource. Geographers study population through **three questions**: how many people there are (**size**), where they live (**distribution**), and how the population changes (**growth and processes**). India's data come from the **Census**, held every **10 years**.

THE CENSUS

The **census** is an official enumeration of population done periodically. The first census in India was in **1872**; the first complete census was in **1881**. Since then it has been held every ten years. The figures below are from the **2011 Census**.

Size and distribution

- On **1 March 2011**, India's population was **1,210.6 million** (about 1.21 billion), about **17.5%** of the world's population, while India covers only **2.4%** of the world's area.
- **Uttar Pradesh** is the most populous state (about **199 million**, about 16% of India's population). Sikkim has a very small population; **Lakshadweep** has only about 64,429 people.
- Almost **half** of India's population lives in just **five states**: Uttar Pradesh, Maharashtra, Bihar, West Bengal and Andhra Pradesh. **Rajasthan**, the biggest state by area, has only about 5.5% of the population.

Population density is the number of persons per unit area (per sq km). India's density in 2011 was **382 persons per sq km**.

- Highest: **Bihar (1,102)**; lowest: **Arunachal Pradesh (17)**.
- **Low density** (below 250 persons per sq km): rugged terrain and unfavourable climate keep population sparse, e.g. in Arunachal Pradesh and other hilly areas.
- **Moderate density**: Assam and most Peninsular states, with hilly, dissected terrain and moderate rainfall.
- **High density**: the Northern plains and **Kerala** in the south have high to very high densities because of flat plains with fertile soils and abundant rainfall.

BIGGEST AREA IS NOT BIGGEST POPULATION

Rajasthan is the largest state by area, but Uttar Pradesh has the largest population. Density compares people to area.

Population growth

Population growth is the change in the number of inhabitants of a country or territory during a specific period.

- **Absolute increase** = later population minus earlier population.
- **Rate of growth** (annual growth rate) is studied in per cent per annum, e.g. 2% per annum means 2 persons added per 100 people per year.

Three phases:

1. **1901–1951:** slow growth; population rose from 238 million to 361 million. Both birth and death rates were high.
2. **1951–1981: population explosion**: the rate of growth was rising; death rate fell rapidly while birth rate remained high. Population rose to 683 million by 1981.
3. **Since 1981:** the rate of growth has started **declining gradually** because birth rates fell. Yet, as the base is huge, the absolute numbers added each year are still enormous (about 182 million were added between 2001 and 2011).

WHY DO ABSOLUTE NUMBERS KEEP RISING?

Even if the growth rate falls from 2% to 1.6%, 1.6% of 1,200 million (19.2 million) is still more than 2% of 600 million (12 million). A large base means big additions every year.

Processes of change

Three main processes change population: **birth rates, death rates and migration**.

- **Birth rate:** number of live births per **thousand** persons in a year. Main cause of growth because India's birth rates have always been high.
- **Death rate:** number of deaths per thousand persons in a year. The rapid **fall in death rate** has been the main cause of rapid growth since 1951 (better health, sanitation, control of epidemics, better food supply).
- **Natural growth rate** = birth rate minus death rate.
- **Migration:** movement of people across regions and territories. It can be **internal** (within the country) or **international**. Internal migration does not change the size of the population, but it changes its **distribution**. In India most migration is **rural to urban** because of **push factors** (poverty and unemployment in villages) and **pull factors** (jobs and better living conditions in cities). Urban population increased from 17.29% in 1951 to **31.80% in 2011**. The number of "million plus cities" rose to **53** in 2011.

Population composition

Age composition: people are grouped as

- **Children** (below 15 years): economically unproductive, need food, clothing, education and health care.
- **Working age** (15–59 years): economically productive and biologically reproductive.

- **Aged** (above 59 years): may be economically productive but are mostly retired.
  A high proportion of children and aged means a high **dependency** on the working population.

Sex ratio: the number of **females per 1,000 males**. India's sex ratio in 2011 was **940** (it has always been unfavourable to females). **Kerala** has the highest sex ratio (**1,084**), and Puducherry 1,038. Delhi has 866 and Haryana only 877.

Literacy: a person aged **7 years and above** who can read and write with understanding in any language is considered literate. Literacy rate in 2011: **74.04%**; males **82.14%**, females **65.46%**.

Occupational structure: activities are grouped as **primary** (agriculture, animal husbandry, forestry, fishing, mining), **secondary** (manufacturing, building and construction) and **tertiary** (transport, communication, commerce, administration, services). In developed countries a high proportion work in secondary and tertiary sectors; in India about 64% still work in agriculture.

Health: life expectancy rose from **36.7 years in 1951 to 67.9 years in 2011**; death rate fell from 25 to about 7.2 per thousand.

Adolescent population: people aged **10 to 19 years**, about one-fifth of India's population. They need good nutrition, especially adolescent girls; many suffer from anaemia.

National Population Policy

India began a comprehensive **Family Planning Programme in 1952**. The **National Population Policy (NPP) 2000** provides a policy framework for:

- free and compulsory school education up to 14 years of age;
- reducing infant mortality rate to below **30 per 1,000 live births**;
- universal immunisation of children against all vaccine-preventable diseases;
- promoting delayed marriage for girls;

- making family welfare a people-centred programme.

NPP 2000 and adolescents: it identified adolescents as a group needing attention: protection from unwanted pregnancies and sexually transmitted diseases, encouraging delayed marriage and child-bearing, education about risky sexual behaviour, providing food supplements and nutritional services, and strengthening legal measures to prevent child marriage.

BOARD FAVOURITE

"What are the three processes of population change?" and "Explain the significance of the age composition of a population" are common. Always define birth rate and death rate "per thousand persons in a year".

Quick recap

- 2011: 1,210.6 million people, 17.5% of the world's population on 2.4% of the area.
- Density 382 per sq km; Bihar highest (1,102), Arunachal lowest (17).
- Growth: slow (1901–51), explosion (1951–81), declining rate (after 1981).
- Processes of change: birth rate, death rate, migration (rural to urban).
- Sex ratio 940; Kerala highest. Literacy 74.04%. NPP 2000 targets health, education and adolescents.

Practice questions

1. According to the 2011 Census, India's population density was

   (a) 324 persons per sq km (b) 382 persons per sq km (c) 450 persons per sq km (d) 1,102 persons per sq km

2. Which state has the lowest population density in India?

   (a) Rajasthan (b) Sikkim (c) Mizoram (d) Arunachal Pradesh

3. Sex ratio is defined as the number of

   (a) males per 1,000 females (b) females per 1,000 males (c) children per 1,000 adults (d) births per 1,000 persons

4. The main cause of India's rapid population growth after 1951 was

   (a) a sharp rise in birth rates (b) large-scale immigration (c) a rapid decline in death rates (d) an increase in literacy

5. Internal migration within a country changes

   (a) the size of the population (b) the distribution of population (c) the birth rate only (d) the sex ratio of the whole country only

6. Which state has the largest population according to the 2011 Census?

   (a) Uttar Pradesh (b) Maharashtra (c) Bihar (d) Rajasthan

7. A person is considered literate in the census if he or she is aged

   (a) 5 years and above and can sign their name (b) 10 years and above and has attended school (c) 7 years and above and can read and write with understanding in any language (d) 15 years and above and can read English

8. What are the three main processes of change of population? Explain each briefly.

9. What are the major features of the National Population Policy 2000? How does it address the needs of adolescents?

Answers: 1. b 2. d 3. b 4. c 5. b 6. a 7. c 8. see notes 9. see notes